

IBM Data Engineering Course | Coursera

BI Dashboards

Summary

Module one

IBM Cognos analytics for data analysis and visualization

Data Visualization and Dashboards with Cognos Analytics

Lesson 1: Introduction to Analytics and Business Intelligence (BI) Tools

By the end of this lesson, you will be able to:

- Define **Analytics**
 - Understand the role of **BI tools** in transforming analytics
 - Identify the **purpose** of BI & Analytics tools
 - List key BI tools in the market today
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What is Analytics?

Analytics is the systematic collection and breakdown of data, statistics, and operations research to build models for improved decision-making.

Key Enablers:

- Growth in **compute power, storage, and memory**
 - Affordable access to **large-scale data processing**
 - Near real-time results thanks to **modern processing capabilities**
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Types of Analytics

Type	Description
Descriptive	Understand what happened in the past
Predictive	Forecast what might happen in the future
Prescriptive	Recommend what actions should be taken to achieve a specific outcome

What Are BI Tools?

Business Intelligence (BI) tools are advanced software applications designed for data preparation, mining, management, and visualization.

Purpose of BI Tools:

- Detect **what is happening** and **why it's happening**
 - Leverage **machine learning** and **statistical models**
 - Empower **descriptive, diagnostic, predictive, and prescriptive** analytics
 - Enable **dashboards, reporting, and self-service** analytics
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Major BI Tools in the Market

Tool	Highlight
IBM Cognos Analytics	AI-powered, includes Watson, natural language processing
Microsoft Power BI	Secure, interactive insights
Tableau	Strong in visualization and usability
Oracle Analytics Cloud	Known for conversational analytics
SAP Business Objects	Intelligent enterprise reporting
TIBCO Spotfire	Scalable for small to enterprise environments

IBM's Investment in Analytics

- IBM has invested **\$25+ billion since 2005** into growing its analytics ecosystem.
 - **Cognos Analytics** covers the **entire data lifecycle** from exploration to operationalization, enhanced with **AI capabilities**.
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Key Takeaways

- Analytics helps build models from data to make better decisions.
 - BI tools transform raw data into actionable insights using modern tech and AI.
 - Cognos Analytics is a top-tier BI solution backed by IBM's vast experience in data analytics.
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Lesson 2: Introduction to IBM Cognos Analytics & Sign-Up Process

Learning Objectives

By the end of this lesson, you will be able to:

- Describe the **IBM Cognos Analytics** platform
 - Understand the **sign-up process** for the 30-day trial version
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What is IBM Cognos Analytics?

IBM Cognos Analytics is a powerful, cloud-based Business Intelligence (BI) and data visualization platform developed by IBM.

Key Features:

- **Data modeling** tools
- **Data exploration & filtering** features
- **Natural Language Querying** via AI-based assistant
- Creation of **interactive dashboards**
- Custom **reports tailored to users**
- Intuitive **drag-and-drop UI**

 The course focuses on using these features to build **insightful dashboards**.

How to Sign Up for the Free Trial

Access the Sign-Up Page:

Go to: ibm.com/products/cognos-analytics

Click: **Start free for 30 days**

Registration Steps:

1. **Fill the form** with:
 - Email (used as IBM ID)
 - First and last name
 - Password
2. Provide additional details:
 - Country and region
 - Phone number
 - Indicate if you're a student
 - Industry sector
 - Closest **data center**

3. Click **Next**
→ IBM sends a **7-digit verification code** to your email
 4. Enter the code → Click **Start Today**
 5. You are redirected to the **Products Dashboard**
 6. Select: **IBM Cognos Analytics on Cloud Trial**
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You Are Now Ready!

 You can now use **Cognos Analytics** to:

- Explore your datasets
 - Build visualizations
 - Create stunning, interactive dashboards
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Key Takeaways

- **Cognos Analytics** is a comprehensive BI platform with visualization, AI, and reporting tools.
 - A **30-day cloud-based trial** is available for hands-on learning and dashboard creation.
 - The registration process is straightforward and grants full access to IBM's analytics environment.
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Lesson 3: Navigating IBM Cognos Analytics & Creating Your First Dashboard

Learning Objectives

By the end of this lesson, you will be able to:

- Navigate the **Cognos Analytics interface**
 - **Upload a CSV** file as a dataset
 - Start building a **dashboard** from a selected template
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Navigating the Interface

Main UI Areas

Section	Functionality
Top Left (Main Menu)	Access to create new items, upload data, browse content, see recent files
Top Right Toolbar	Trial countdown, upgrade options, access to AI assistant , settings
Homepage	Quick access to: tutorials, product tour, content creation, and recently used items

Pro Tip:

Use the **search bar** or the **AI Assistant** to ask data-related questions and discover insights faster.

Uploading a CSV Dataset

1. Go to **Main Menu** → select **Upload Data**
2. Select your `.csv` file → Click **Open**
3. Observe progress in the status bar (you can expand details)
4. Upon completion, your file appears in **My content**

 Uploaded data can be:

- Moved to **Team content** (shared)
 - Renamed or deleted
 - Reused in multiple dashboards
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Creating a Dashboard

Steps:

1. Select your **uploaded dataset**
2. Click **Create Dashboard**
3. Choose a **layout template**

Example: 4 small panels + 1 large panel

4. Click **Create** to launch the dashboard editor
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Dashboard Interface Elements

Icon	Function
Sources	View fields from your selected dataset
Pins	Visualizations you have “pinned” for reuse
Visualizations	Access system-defined charts (e.g., bar, pie, tree maps); upload custom visuals
Widgets	Add text, images, shapes, videos, audio, or embed web content

Pinning Visualizations

- Add a chart (e.g., Decision Tree) to the canvas
- Click **Pin** on the top toolbar
- Confirm pinning from the popup
- Reuse it later via the **Pins** panel

Key Takeaways

- The **Cognos UI** is user-friendly and modular
- You can **upload CSVs** to populate datasets quickly
- Dashboards are built using a **template-first** approach
- Visualizations can be **pinned** for faster reuse
- **Widgets** add rich media and interactive content to dashboards

Pro Dashboarder Tip:

Use the **AI Assistant** at any time to explore the data intelligently—it’s like having a data scientist at your side (but doesn’t ask for coffee breaks ☕).

Lesson 4: Creating a Simple Dashboard in IBM Cognos Analytics

Learning Objectives

By the end of this lesson, you will be able to:

- Build a **simple interactive dashboard**
 - Create visualizations in **three different ways**
 - Apply and configure **interactive filters**
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Step-by-Step: Building a Dashboard

1. Select Dataset

- Go to **My Content**
- Choose your uploaded dataset
- Click **Create > Dashboard**
- Select a **layout template** (e.g., 4 top panels + 1 bottom panel)

2. Adjust Data Field Properties

Example: Set `Order Number` field usage from "Measure" → "Identifier" if it's a categorical field.

3 Methods to Create Visualizations

Method 1: Drag Data Fields onto Canvas

- Drag `Sales` → top-left panel → Cognos auto-creates a chart
- Drag `Order ID` → summary table (you can change visualization type if needed)
- Drag `Quantity Ordered` → another panel
- Drag `Sales` again → set summarization to **Average** in a fourth panel

Result: You now have **Key Performance Indicators (KPIs)** displayed.

Method 2: Select Visualization Type First

- Choose a specific chart (e.g., **Map** visualization)
- Drag `Country` → **Location**
- Drag `Sales` → **Color by**
- Resize and customize layout with % width indicators

Chart Properties include: **Legend, Layers, Regions, Coordinates**

Method 3: Use the Assistant

- Type a question like: *"Which deal size has the most sales?"*
- Receive **automatic visualizations** and recommended alternatives (bar, scatter, bubble)
- Drag desired chart onto canvas → Resize as needed

Interactivity Features

Click-to-Filter

Clicking elements in a chart (e.g., "Large" deal size) updates all other visuals **in real time**.

Adding Filters

- Drag a field like `Status` into "This tab" or "All tabs"
- Filter data (e.g., only show **On-Hold** orders)
- All charts dynamically reflect the filtered dataset

Insight: "On-Hold" orders are only present in a few countries — good operational sign!

Key Takeaways

- You can choose from **various dashboard templates**
 - **Field properties** can be adjusted for better categorization
 - Visuals can be created by:
 - Dragging fields
 - Selecting chart types
 - Using the **AI Assistant**
 - Cognos dashboards are inherently **interactive**
 - Filters and chart interactions provide **real-time updates**
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 **Pro Insight:** Cognos lets you build dashboards the way you think—drag & drop, guided AI, or manual control. It's like the Swiss Army knife of BI tools.

Lesson 5: Advanced Dashboard Capabilities in IBM Cognos Analytics

By the end of this lesson, you will be able to:

- Create **custom calculations** in dashboards
- Implement **navigation paths** for drill-down analysis
- Use **exclude filters** and **top/bottom filters**
- Enhance dashboards with **infographics**



1. Creating Calculations

Similar to Excel formulas, you can build custom fields using attributes + operators.



Steps:

1. Right-click your dataset in **Data Source Pane** → Select **Create Calculation**
2. Name the calculation: e.g., Margin
3. Build formula:

`[MSRP] - [Sale Price]`

4. Use drag-and-drop or type attribute names (Cognos auto-suggests)
5. Result: New calculated field appears like any other data item



Use case: Visualizing `Margin per Product Line` to identify profit/loss segments



Example insight: Product line "**Trains**" shows a **negative margin**



2. Creating Navigation Paths (Drill-down)



Purpose:

Explore data **hierarchically** (e.g., Product → Customer → Order)



Steps:

1. In Data Source Pane, right-click → **Create Navigation Path**
2. Define the drill-down order (e.g., `Product Line > Customer Name > Order ID`)
3. In dashboard:
 - Click on an item (e.g., “Trains”) → **Drill down**
 - Analyze deeper trends and responsible entities (e.g., Mini Gifts Co.)

To **drill up**, use the UI's back arrow or context menu.



3. Excluding Data from Visualizations



Steps:

1. After adding fields like `Sales Status + Product Line` to canvas

2. Right-click on an unwanted item (e.g., “Shipped”) → **Exclude**
3. Visualization updates to focus on remaining data (e.g., statuses other than “Shipped”)

 Insight: Great for decluttering charts and zeroing in on pain points.

4. Top/Bottom Filters

Highlight most or least impactful items in large datasets.

Steps:

1. Add attributes like `Sales + Customer Name` to a visualization
2. Right-click on `Sales` → **Top or Bottom**
3. Choose the number of results (default = **Top 10**)

 Use case: Spot your **Top 10 customers by revenue**

5. Generating Infographics

Make your dashboards pop with visual appeal!

Steps:

1. Go to **Widgets Menu**
2. Choose a shape/icon (e.g., **Piggy Bank** )
3. Drag & drop it over your target chart (e.g., sales graph)

 Result: An engaging, branded, and **memorable** data visualization

Key Takeaways

Feature	Description
Calculations	Create custom metrics with arithmetic/logic
Navigation Paths	Enable drill-down for hierarchical data exploration
Exclude Filter	Remove unwanted values from visualizations
Top/Bottom Filter	Limit charts to most/least impactful entities
Infographics	Add visual icons for intuitive storytelling

🔥 **Power Tip:** Combining filters, calculations, and drill-downs transforms your dashboards from "meh" to **mind-blowing**. Cognos doesn't just visualize data — it tells a story!

📘 Lesson 6: Accessing Your Data Warehouse Using IBM Cognos Analytics

🎯 Learning Objectives

By the end of this lesson, you will be able to:

- Connect Cognos Analytics to a **database or data warehouse**
 - Build a **dashboard using live warehouse data**
 - Add and configure **multiple visualizations** in a dynamic dashboard
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📁 1. Supported Data Sources

Cognos Analytics supports cloud and on-premise sources like:

- **IBM Db2 Warehouse, Amazon Redshift, Oracle**
 - **Microsoft SQL Server, MySQL, PostgreSQL, Snowflake**
 - **MongoDB** and others
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🔗 2. Step-by-Step: Connecting a Data Warehouse

✅ Overview of Key Steps:

Step	Action
1.	Identify database, schema, tables, and get credentials
2.	In Cognos, go to Manage > Data Server Connections
3.	Select warehouse type (e.g., IBM Db2), use JDBC URL with credentials
4.	Test connection → ✅ Success
5.	Load metadata from the schema
6.	Create a Data Module from selected tables
7.	Use that module to build a dashboard



3. Creating a Data Module

1. Navigate to the newly added connection (e.g., `MyDB2`)
2. Select the schema and table (e.g., `CustomerLoyalty`)
3. Click to open and preview the data
4. Save the module (e.g., **MyCustomerLoyaltyModule**) in **My Content**

A **Data Module** acts as a link or container to your tables, enabling analysis and dashboarding.



4. Building the Dashboard



Dashboard Setup

- Go to **New > Dashboard**
- Choose layout: **Tabbed** or **Infographic**
- Add your **data module** as the source



Adding Visualizations

Attribute	Visualization Type	Description
Quantity Sold	KPI or Summary	Shows total units sold
Product Line over Time	Bar Plot	Visualizes sales trends by category
Country + Quantity	World Map	Regional sales, interactive selection



Interactive Behavior

- Click a region → other charts **update dynamically**
 - Resize, format, rename titles as needed
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5. Sharing & Saving

- Save the dashboard with a relevant name
 - Cognos provides a **shareable link** to view dashboards in the browser
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Key Takeaways

Concept	Description
Data Server Connection	Establishes a secure connection to a warehouse/database
Data Module	A reusable container for specific tables/fields
Interactive Dashboard	Combines multiple visualizations from a module with filter & drill-down
Templates	Various layout options for storytelling & design

📌 **Pro Tip:** Think of `Data Server Connection` as your **plug**, and `Data Module` as your **power strip**—you can connect multiple visualizations from the same source effortlessly ⚡.

Lesson's summary:

Analytics is the building of data models to make better decisions, and business intelligence is a technology that enables data preparation, mining, and visualization. You learned that accessing a database in Cognos requires a data server connection and a data module. You also learned how to add a data module to a dashboard and use its fields and columns to create visualizations. And you gained practical hands-on experience with IBM Cognos Analytics to create dashboards and visualizations.

Module Two

Data visualization and Dashboards with google Looker Studio

Creating and configuring visualizations with google Looker Studio\

Google Looker Studio – Lesson 1: Getting Started with Google Looker Studio

Learning Objectives

By the end of this lesson, you will be able to:

- Define what **Google Looker Studio** is
 - Understand how to **create and edit reports**
 - Explore **data source connection options**
 - Share reports using different **collaboration methods**
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What is Google Looker Studio?

Google Looker Studio (formerly *Google Data Studio*) is a **free, cloud-based** data visualization tool that allows you to build interactive reports and dashboards.



Key Features:

- Create multi-page, visually rich **reports**
 - Use data from **multiple sources**
 - Share insights easily with **teams or clients**
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Accessing the Tool

- Go to lookerstudio.google.com
 - Sign in with any valid **Google account**
 - You'll land on the **homepage**, which includes:
 - **Recent reports**
 - **Templates gallery**
 - **Data sources** and **explorer** tabs
-



Two Ways to Create a Report



1. From a Template

- Navigate to **Templates**
- Choose one (e.g., for Google Analytics, Sheets, BigQuery)
- Click **Edit and Share** → A copy is saved to **Recent**
- Enter **Edit Mode** to modify text, data, and charts



2. From Scratch

- Click **Create > Report**
 - Select a **connector** for your data (e.g., upload a file)
 - Choose the file or drag it into the upload area
 - Click **Add to Report** after processing completes
-



The Report Editor

- Use **Edit Mode** to change visualizations, titles, layout
- Add new fields, metrics, or dimensions

- Use **View Mode** to preview the report as a reader

Sharing Options

Method	Description
Email Invite	Share with individual users or Google Groups
Link Sharing	Copy link for access (can be restricted or public)
Schedule Delivery	Set up recurring email delivery of reports (e.g., daily, weekly)
Embed	Add report into websites or apps (as HTML/URL)
Download as PDF	Export report for offline viewing or attachment

Key Takeaways

- Looker Studio is ideal for **data storytelling** using cloud-powered dashboards
 - Reports can be created from **templates** or **blank canvases**
 - You can **upload your own datasets** or connect to external data sources
 - Sharing is flexible and encourages **real-time collaboration**
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✦ **Pro Tip:** If you're familiar with Google Sheets and Slides, you'll feel right at home—Looker Studio combines both worlds with a data twist!

Lesson 2: Connecting Data, Visualizing Reports & Customizing Layouts

Learning Objectives

By the end of this lesson, you will be able to:

- Understand the difference between **connectors** and **data sources**
 - Add charts and **visualizations** to reports
 - Configure **themes** and **layout settings**
-

Understanding Connectors vs. Data Sources

Term	Definition
Connector	A tool that links Looker Studio to an external data source (e.g., Google Sheets, SQL DB)
Data Source	A configured instance of a connector with specific schema/table selected

 Example:

- Connector = Google Sheets
- Data Source = Sheet named “Sales_2024” from your Drive

Adding Data to a Report

Option 1: New Data Source

- Go to “**Add data to report**” → **Connect to data**
- Choose a connector (e.g., Google Sheets, BigQuery)
- Select/upload your file or connect to external DB
- Result: Report is created with the data **embedded**

Option 2: Existing Data Source

- Go to “**My data sources**”
- Pick a previously connected data source
- Report opens with first chart auto-generated

Adding & Configuring Visualizations

Chart Types Available:

- Bar Chart
 - Pie Chart
 - Line Chart
 - Map
 - Bubble Chart
- Add via “**Add a chart**” from toolbar

Chart Properties Pane

Tab	Description
Set-up	Configure data source, dimension, metrics, filters
Style	Visual styling: colors, background, legend, data labels, etc.

Tab	Description
Data Pane	Shows all fields available in the connected data source

 **Pro Tip:**
You can **drag fields** from the data pane directly into the **dimension/metric** section to instantly update your chart.

Example changes:

- Dimension → from `First Name` → `Product Line`
- Metric → from `Record Count` → `Revenue`

Themes and Layouts

Unlike IBM Cognos, Looker Studio doesn't use fixed dashboard panels—you **have full layout freedom** on the canvas.

Theme & Layout Pane

- Access from **File Menu** or **Toolbar**
- **Theme Tab:**
 - Choose predefined color themes
 - Customize fonts, backgrounds, chart colors
- **Layout Tab:**
 - Page navigation (top, side, hidden)
 - Canvas size
 - Display mode (fit to width, actual size)
 - Show/hide guides and grid

Key Takeaways

Feature	Description
Connector	Connects to external data platform
Data Source	A specific data instance used in the report
Chart Properties	Customize charts' data and visual appearance via properties panel
Drag & Drop Fields	Modify dimensions/metrics with ease
Themes & Layout	Customize the report's entire visual identity and layout configuration

 **Pro Tip:** Looker Studio is like a blank canvas — you're the artist! Mix charts, maps, and colors freely to build a masterpiece of insight.

Lesson 3: Working with Pages and Blended Data

Learning Objectives

- Configure **page settings** like canvas size and background
 - Add and **navigate multiple pages** within a report
 - **Blend data** from multiple tables using relationships
-

1. Configuring Page Settings

How to Access Page Settings:

- Right-click on **blank canvas** → `Current page settings`
- OR use the **Page menu** from the top ribbon

Available Configurations (Style Tab):

Setting	Options
Canvas Background	Choose solid colors
Canvas Size	Auto-size, Preset dimensions, Custom

 Tip: Use **Auto-size** if you're unsure how many visualizations will be added

2. Managing Multiple Pages

Adding Pages:

- Use the **Page menu** → `New page`
- Or click + in the **Report Pages pane**
- Shortcut: `Ctrl + M`

Navigating Pages:

- Use **navigation arrows** at top
- Click **page number dots** (left side)
- Use the **Report Pages pane** on the right

Renaming Pages:

- Click the **three dots** (...) next to the page → Rename

3. Blending Data from Multiple Tables

Use **Blend Data** when your data lives in more than one table but shares a common field.

Steps to Blend:

1. Upload & add both datasets to the report
2. Add a chart (e.g., column chart) to the canvas
3. In the chart's properties pane → Click **Blend Data**
4. In the Blend Data window:
 - Choose both tables
 - Select **join key** (e.g., `cust_ID`)
 - Choose **join type**: Left Outer Join
 - Add other dimensions like Gender
5. Save the configuration

Result:

- A **single chart** that pulls and visualizes data from **multiple tables**
- The **data source name** will update to show it's blended

Key Takeaways

Feature	Description
Page Settings	Customize canvas size, background, and layout for each page
Multi-Page Reports	Use multiple pages to organize complex dashboards
Blended Data	Combine two+ data sources using a shared field and join type
Chart Flexibility	Charts can now visualize merged data , enhancing insight potential

 **Pro Tip:** Use blended data to **avoid exporting & merging manually**—it saves time and keeps your dashboard dynamic and scalable!  

Lesson 4: Adding Controls, Filters & Visualizations

Learning Objectives

By the end of this lesson, you will be able to:

- Add **controls** to make reports interactive
 - Apply **filters** to manipulate visible data
 - Use various **visualization types** to enrich your reports
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1. Adding Controls to Reports

Controls let users **interact with data**—either by filtering it or providing input.

Types of Controls

Type	Purpose
Drop-down list	Select predefined values
Slider	Adjust numeric or date ranges
Input box	Enter free text or numeric values
Date range picker	Filter by time intervals
Button control	Trigger navigation or actions

 Add controls via:
→ **Toolbar > Add a control**

2. Adding Filters to Charts

Filters allow you to **focus** or **exclude** specific data values.

How to Add a Filter:

1. Select a **chart**
2. Go to the chart's **properties pane** → `Create Filter`
3. Define filter:
 - Include/Exclude condition
 - Set **field**, **operator**, and **value**
 - (Optional) Add **AND/OR** logic

 Example:
Exclude data where `City = Vancouver` → Chart updates immediately



3. Adding and Configuring Visualizations

Looker Studio supports a **wide range of charts**, from standard to advanced.



Common Charts:

- Bar chart
- Line chart
- Pie chart
- Table
- Scorecard
- Map



Advanced Charts:

- Area chart
- Pivot table
- Tree map
- Heat map
- Gauge
- **Waterfall chart**



Configuration via Chart Properties Pane:

Tab	Purpose
Setup	Set data source, dimension, metrics, filters
Style	Customize design: color, labels, legends, borders

→ Drag fields directly from **Data Pane** into dimensions/metrics to update chart



View your dashboard in **Read-Only Mode** with the “View” button



Map elements are zoomable and draggable in both Edit and View modes



Visualization Examples

Chart Type	Example Use Case
Tree Map	Hierarchical category relationships
Heat Map	Intensity or frequency of values
Waterfall	Sequential contribution to totals

Key Takeaways

Feature	Purpose
Controls	Add interactivity (drop-downs, sliders, input fields)
Filters	Exclude/include data from charts based on logic
Visualizations	Use diverse chart types to communicate insights visually
Customization	Fine-tune charts via Setup + Style tabs in the properties pane

📌 **Pro Tip:** Don't just present data — let users *interact* with it! Controls and filters empower your audience to explore insights dynamically. ✨

Lesson's Summary:

- The fundamentals of the Looker Studio application.
 - How to navigate around the Looker Studio user interface.
 - How to create reports in Looker Studio.
 - How to connect to data sources in Looker Studio.
 - How to create dashboards in Looker Studio.
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Course Summary

- **IBM Cognos Analytics** offers sophisticated tools for creating interactive dashboards and comprehensive data visualizations, from initial data uploading to employing advanced functionalities like creating calculations and custom visualizations.
- **Google Looker Studio** enables dynamic and interactive data visualizations, guiding you through the essentials from setting up and connecting to data sources to creating detailed reports and configuring advanced dashboard settings with features like calculated fields and dynamic filtering.
- **Hands-on labs and quizzes** throughout both modules ensured practical experience and understanding, allowing you to apply the concepts taught to real-world data scenarios effectively.

- **Operational and strategic uses of BI tools** are now within your capabilities, enabling the creation and manipulation of powerful visualizations and making informed decisions using real-time data analysis.
- **Selection and application of BI tools** for specific business needs and complex problem solving are skills you have acquired, equipped with the ability to use advanced analytical techniques.

This course has prepared you for roles in data analytics, business intelligence, and decision-making, enhancing your ability to harness the power of BI tools in professional scenarios.